teetered along an extremely narrow ledge. A tiny deviation in conditions billions of years ago would have led to a present-day universe very different from the one revealed by astronomers' measurements. This is known as the *flatness problem.*

Although we've covered the essential idea, it's important to understand the sense in which the flatness problem is a problem. By no means does the flatness problem show that the standard big bang model is wrong. A staunch believer reacts to the flatness problem with a shrug of the shoulders and the curt reply "That's just how it was back then," taking the finely tuned matter/energy density of the early universe—which the standard big bang requires to yield predictions that are in the same ball-park as observations—as an unexplained given. But this answer makes most physicists recoil. Physicists feel that a theory is grossly unnatural if its success hinges on extremely precise tunings of features for which we lack a fundamental explanation. Without supplying a reason for why the matter/energy density of the early universe would have been so finely tuned to an acceptable value, many physicists have found the standard big bang model highly contrived. Thus, the flatness problem highlights the extreme sensitivity of the standard big bang model to conditions in the remote past of which we know very little; it shows how the theory must assume the universe was *just so*, in order to work.

By contrast, physicists long for theories whose predictions are insensitive to unknown quantities such as how things were a long time ago. Such theories feel robust and natural because their predictions don't depend delicately on details that are hard, or perhaps even impossible, to determine directly. This is the kind of theory provided by inflationary cosmology, and its solution to the flatness problem illustrates why.

The essential observation is that whereas attractive gravity amplifies any deviation from the critical matter/energy density, the repulsive gravity of the inflationary theory does the opposite: it *reduces* any deviation from the critical density. To get a feel for why this is the case, it's easiest to use the tight connection between the universe's matter/energy density and its curvature to reason geometrically. In particular, notice that even if the shape of the universe were significantly curved early on, after inflationary expansion a portion of space large